Empirical–generated learning-curve comparisons
Shared observed subset-size ranges

A BRCA miRNA-seq
TC · cVAE 1:5
No AE head
Mean source ARI: 0.0055

B HNSC miRNA-seq
raw · cVAE 1:50
No AE head
Mean source ARI: 0.0058

C HNSC miRNA-seq
TC · cVAE 1:50
No AE head
Mean source ARI: 0.0080

D COAD RNA-seq
TC · cVAE 1:50
AE head (2 iterations)
Mean source ARI: 0.00014

	A	B	C	D
LR Macro-F1	B +0.073 D 0.073	B +0.007 D 0.019	B +0.146 D 0.146	B +0.119 D 0.119
SVM Macro-F1	B -0.002 D 0.018	B +0.005 D 0.015	B +0.147 D 0.147	B +0.077 D 0.080
KNN Macro-F1	B +0.001 D 0.009	B +0.001 D 0.020	B +0.159 D 0.159	B +0.069 D 0.079
RF Macro-F1	B -0.006 D 0.009	B -0.007 D 0.017	B +0.134 D 0.134	B +0.054 D 0.054
XGB Macro-F1	B +0.005 D 0.016	B -0.033 D 0.040	B +0.096 D 0.096	B +0.058 D 0.063

Macro-F1: 0.00, 0.25, 0.50, 0.75, 1.00
Candidate subset size: 250, 500, 750, 1000 (A); 150, 300, 450 (B, C, D)

Empirical Generated

Points: observed means ±1 draw SD. Lines/bands: archived fit / pointwise 95% fitted-mean interval.
Each column uses its shared observed range; x-axis ranges differ between tasks.
B: signed average gap; D: absolute average gap. Both use shared observed means. ARI: mean of five fidelity draws.